Week 2 Tutorial Questions & Notes

- Definite Integral using the limit definition

Formula

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x$$

- $\sum$: sum
- n: no. of rectangles
- $f(x_i)$: height
- Δx: width
- $n \to \infty$: limit to infinity
- $i=1$: starts at 1

} Area under a region

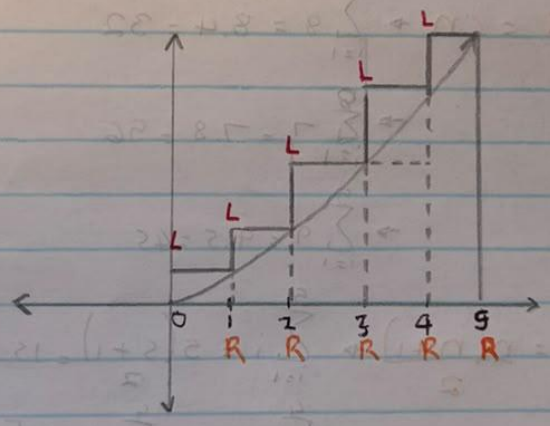
Right-endpoint → Over-estimation **Left-endpoint** → Under-estimation

$$\Delta x = \frac{b-a}{n}$$

~~x_i = at~~

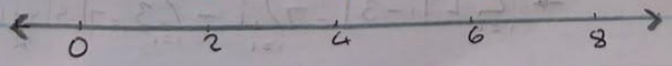
$x_i = a + \Delta x\, i$ (a: first value)

$x_i = a + \Delta x(i-1)$



Example → $f(x) = x^3$ $[0;8]$ (a, b)

x_1 x_2 x_3 x_4



$x_i = a + \Delta x_i$

$x_1 = 0 + 2(1) = 2 \mid x_2 = 0 + 2(2) = 4 \mid x_3 = 0 + 2(3) = 6 \mid x_4 = 0 + 2(4) = 8$

$$A = \lim_{x\to\infty} \sum_{i=1}^{n} f(x_i)\,\Delta x \quad [0;8] \quad f(x) = x^3$$

$\Delta x = \frac{b-a}{n} = \frac{8-0}{n} = \frac{8}{n}$ → find Δx

$x_i = a + \Delta x_i$ → find x_i using right-endpoints
$= 0 + \frac{8}{n}i$

$A = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i) \cdot \Delta x$ → $A = \lim_{n\to\infty} \sum_{i=1}^{n} f\left(\frac{8i}{n}\right) \cdot \left(\frac{8}{n}\right)$

$A = \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{8i}{n}\right]^3 \left[\frac{8}{n}\right]$ → sub the o.g function (x^3) in.

$A = \lim_{n\to\infty} \sum_{i=1}^{n} \left[\frac{512i^3}{n^3}\right]\left[\frac{8}{n}\right] = \lim_{n\to\infty} \sum_{i=1}^{n} \frac{4096i^3}{n^4}$ → Multiply everything

•Summation formulas (4)

① Constant: $\sum_{i=1}^{n} c = cn$ → $\sum_{i=1}^{n} 8 = 8 \cdot 4 = 32$

→ $\sum_{i=1}^{8} 7 = 7 \cdot 8 = 56$

→ $\sum_{i=1}^{5} 9 = 9 \cdot 5 = 45$

② Variable: $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$ → $\sum_{i=1}^{5} i = \frac{5(5+1)}{2} = 15 = 1+2+3+4+5$

→ $\sum_{i=1}^{4} 6i = 6\sum_{i=1}^{4} i = 6\left[\frac{4(4+1)}{2}\right] = 6[10] = 60$

→ $\sum_{i=1}^{5} [7i - 3] = 7\sum_{i=1}^{5} i - \sum_{i=1}^{5} 3 = 7\left[\frac{5(5+1)}{2}\right] - (3 \cdot 5)$

$= 7[15] - 15$

$= 105 - 15$

$= 90$

$\rightarrow \sum_{i=1}^{8}[9i+7] = 9\sum_{i=1}^{8} i + \sum_{i=1}^{8} 7 = 9\left[\frac{8(8+1)}{2}\right] + (7 \cdot 8)$

$= 9[36] + 56$

$= 380$

③ Variable^2 ; $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6} \rightarrow \sum_{i=1}^{6} i^2 = \frac{6(6+1)(2 \cdot 6+1)}{6} = 7.(12+1)$

$= 7.(13)$

$= 91$

$\rightarrow \sum_{i=1}^{5} i^2 = \frac{5(5+1)(2 \cdot 5+1)}{6} = 6.(10+1)$

$= 6.11$

$= 66$

④ Variable^3 : $\sum_{i=1}^{n} i^3 = \frac{n^2(n+1)^2}{4} \rightarrow \sum_{i=1}^{4} i^3 = \frac{4^2(4+1)^2}{4} = 4(4+1)^2 = 4(5)^2 = 4 \cdot 25 = 100$

$\rightarrow \sum_{i=1}^{5} i[i^2+4i] = \sum_{i=1}^{5} i^3+4i^2$

$\sum_{i=1}^{5} i^3 = \frac{5^2(5+1)^2}{4} = 225$

$\sum_{i=1}^{5} 4i^2 = 4\left[\frac{5(5+1)(2(5)+1)}{6}\right]$

$= 220$

$\therefore \sum_{i=1}^{5} i^3+4i^2 = 225+220 = 445$

Check Again

$\rightarrow \sum_{i=1}^{5} i[i^2+4i] = \sum_{i=1}^{5} i^3+4i^2$

$= \sum_{i=1}^{5} \frac{5(5+1)^2}{4} + 4\sum_{i=1}^{5} 4 \cdot \frac{[5(5+1)(2 \cdot 5+1)]}{6}$

$= 5(6)^2 + 54 \cdot \frac{[5(6)(11)]}{6}$

$= 180 + 20 + 66$

$= 266$

$= 225 + 20 + 55$

$=$

continue Example → A. $\lim_{n\to\infty} \frac{4096}{n^4} \sum_{i=1}^{n} i^3$ → rearrange formula to be able to use summation formulas

$= \lim_{n\to\infty} \frac{4096}{n^4} \cdot \frac{n^2(n+1)^2}{4}$ → sub formula in for i^3

$$= \lim_{n\to\infty} \frac{4096}{n^4} \cdot \frac{n^2(n+1)^2}{4} = \lim_{n\to\infty} \frac{1024}{n^2} \cdot (n+1)^2 = \lim_{n\to\infty} \frac{1024(n+1)^2}{n^2}$$

$$= 1024 \cdot \lim_{n\to\infty} \frac{n^2+2n+1}{n^2} \Big/ \frac{1}{n^2} = 1024 \cdot \lim_{n\to\infty} \frac{1+\frac{2}{n}+\frac{1}{n^2}}{1}$$

$$= 1024 \cdot 1 + \underset{\to 0}{\frac{2}{0}} + \underset{\to 0}{\frac{1}{0^2}} = 1024$$